BIG MODULAR LADDER — S3 schematic: analog version of the modules/16bit polysynth ladder filter. Circuit intent and numbers: docs/research.md

Signal nets are wired; supplies use power symbols. LB0..LB4 (ladder base bias), ENV_BUF (envelope x DEPTH) and VCA_ENV (VCA envelope / CV) are joined by their net labels.

MIDI (TS, JP1 open: 1–2 protected, 2–3 direct) and GATE (threshold 2.2 V):
NPN inverters on +5V; firmware sets PORT_INVEN on PB3 / PB2.

ENVELOPE: ATtiny1616 at 5 V, AR envelope (ENV TIME) x DEPTH on the PA6 DAC (0..4.35 V) -> U4A -> ENV_BUF:
the undepthed envelope on the PB0 PWM (2 x 340 Hz RC) -> U3 NC1 -> VCA_ENV.
CV BYPASS: J3 patched (PA3 high) -> DAC = CV (PA2) x DEPTH; U3 COM1 = CV 0..5 V scaled to 0..4.35 V.
REC (PC2) + LED (PC3): 4-bar ENV TIME / DEPTH recorder on MIDI clock (modules/acid CvRecorder).

VACTROL VCA (Bread Modular V2CA + LOW/SVF driver): I_LED = VCA_ENV / (470R + CURVE);
LDR in series with R49 into U11A: VCA OUT = −FILTER x 10k / (10k + R_LDR).

OUTPUT: U6 followers on the top nodes, U7A x10 difference amp (LAD_DIF = 0.05 x IN),
U9A AC coupled x20.6 -> 1k -> FILTER OUT (IN x 1.03, in phase).

RESONANCE: U11B x −1.42 (LAD_DIF − 0.05 x IN) -> C20 -> vactrol LDR -> feedback base (1k):
k = 14.3 x 1k / (1k + R_LDR): dry level kept at any k = the 16bit (1 + k) makeup.
RES knob + RES CV -> RES_X (passive mix, no CV attenuator) -> U7B: I_LED = RES_X / 330R.

AUDIO IN: U5B x −1. LADDER: 200k / 1k into the driver (+/−25 mV): 4 rungs, 47n NP0;
bias chain LB0..LB4 1.5 V steps.

CUTOFF + DEPTH summer -> 1/40 divider -> expo (Q3A reference at 0.55 mA, Q3B = ladder tail).
D6 clamps V_S at −0.6 V (−31 kHz).

POWER: +12V -> D1 -> +12V_F (op amps, ladder, U1 HT7550-1 -> +5V); −12V -> D2 -> −12V_F.

SUPPLY PINS + DECOUPLING

Circuit intent: docs/research.md: ngspice check: tools/sim_ladder.py
SCHEMATIC S3 — analog 16bit ladder filter (vactrol resonance) + recordable envelope + vactrol VCA

Big Modular	
Sheet: /	
File: ladder.kicad_sch	
Title: Big Modular — LADDER	
Size: A2	Date:
KiCad E.D.A. 10.0.6	Rev: S3
	Id: 1/1